

ITPROFEL2: OPEN SOURCE PROGRAMMING

Learning Module: Midterm - Week 5 (Detailed Code Guide)

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I. Overview and Objectives

This week, we transition from theoretical open-source concepts directly into practical Python implementation. Proper coding habits, syntax mastery, and understanding the core mechanics of variables and operations begin here.

Intended Learning Outcomes (ILOs):

- Set up the local Python environment and IDLE.
- Apply Python's strict indentation rules to define code blocks.
- Declare variables using correct identifiers, data types, and standard naming conventions.
- Execute mathematical and logical operations using Python operators.
- Implement basic Input and Output (I/O) functions to interact with users.

II. Python Syntax: The Power of Indentation

Unlike C or Java which use curly braces {}, Python relies entirely on **indentation** to define blocks of code. A standard indentation is **4 spaces**.

```
# Correct Indentation Example
if 5 > 2:
    print("Five is greater than two!") # Indented with 4 spaces

# INCORRECT Indentation (Will cause an IndentationError)
if 5 > 2:
print("Five is greater than two!")
```

III. Variables, Identifiers, and Data Types

1. Declaring Variables and Identifiers

Identifiers should follow *snake_case* conventions. Python is dynamically typed, meaning you do not declare the type explicitly.

```
# Variable Assignment
student_name = "Juan Dela Cruz"
year_level = 3
is_enrolled = True

print(student_name)
```

2. Dynamic Typing in Action

A variable can change types dynamically during the execution of the program. You can use the built-in `type()` function to check a variable's current data type.

```
data_value = 100
print(type(data_value)) # Output: <class 'int'>

data_value = "One Hundred"
print(type(data_value)) # Output: <class 'str'>
```

3. Code Documentation (Comments)

```
# This is a single-line comment.
# It explains what the following code does.
total_score = 95 # Inline comment explaining the variable

"""
This is a multi-line comment or docstring.
It is often used to describe the purpose of a module, function,
or complex logic block in open-source projects.
"""
```

IV. Operators in Python

Python supports standard arithmetic operators. Pay special attention to Division (`/`), Floor Division (`//`), and Modulo (`%`).

```
a = 15
b = 4

print(a + b) # Addition: 19
print(a - b) # Subtraction: 11
print(a * b) # Multiplication: 60

# Division differences
print(a / b) # Float Division: 3.75
print(a // b) # Floor Division (rounds down): 3
print(a % b) # Modulo (remainder): 3
print(a ** 2) # Exponentiation (15 squared): 225
```

V. Input and Output (I/O) Basics

Use `input()` to get data from the user. Note that `input()` **always** returns a string. If you need to perform math on the input, you must type-cast it using `int()` or `float()`.

```
# Basic Output
print("Welcome to the system.")

# Basic Input (String)
user_name = input("Enter your username: ")
print("Hello, " + user_name + "!") # String concatenation

# Input with Type Conversion for Math
raw_age = input("Enter your age: ")
age_in_five_years = int(raw_age) + 5
print("In 5 years, you will be: ", age_in_five_years)
```

Laboratory Coding Session: Environmental Data Tracker

Scenario: You are building a CLI utility for the "Protect Our Bakawan" project to track mangrove seedlings. This activity will test your ability to capture input, perform type casting, calculate basic math, and format the output.

Instructions:

1. Create a new Python script named `bakawan_tracker.py`.
2. Use `input()` to ask the user for the **Volunteer Name**, **Site Location**, and **Seedlings Planted**.
3. Cast the seedlings input from a string into an integer so you can perform math on it.
4. Calculate an 85% survival rate (multiply the seedlings by `0.85`) and cast the final result back to an integer to avoid showing decimal/partial trees.
5. Use the `print()` function to output a cleanly formatted summary matching the sample run below.

Sample Program Run (Expected Output):

```
Enter Volunteer Name: Henry
Enter Site Location: Mangrove Zone A
Enter seedlings planted: 150

--- BAKAWAN GROWTH REPORT ---
Volunteer: Henry
Location: Mangrove Zone A
Seedlings Planted: 150
Estimated Surviving Trees: 127
```

Submit your completed `.py` file to the instructor once your terminal output matches the format above.